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Stel $\theta = \arctan x$ met $x \in \mathbb{R}$ en $\theta \in \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$, dan is $\tan \theta = x$

$$\begin{aligned}\cos(2 \arctan x) &= \cos(2\theta) = \frac{\cos^2 \theta - \sin^2 \theta}{1} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{\frac{\cos^2 \theta}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\cos^2 \theta}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta}} = \\ \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} &= \frac{1 - x^2}{1 + x^2}\end{aligned}$$

References